

Real Time Network Intrusion Detection using Machine Learning Technique

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Abstract—This paper reflects the work carried out in network security using distinctive machine-based learning techniques. In response the exponential increase in network space breaches and data leaks, the demand for a system that can detect anomalies and notify the system admin is imperative. Using packet sniffing modules, we capture the packets and then compare them to a pre-trained machine learning module trained on the NSL KDD dataset to detect ambiguous packets. By selecting the desired port, the IDS (Intrusion Detection System) sniff all incoming packets and categorizes them as anomalous if their behavior is not normal. On successful prediction, we present the user with a choice to act against the prescribed threat or ignore it as per the user's request. A detailed analysis report shall then be presented periodically to provide an overview of the overall health of the system on which our IDS system has been deployed.

Keywords—packet-sniffing, real time intrusion, decision tree, network packets, NSL KDD, Intrusion Detection System, Anomaly Detection, cyber security, packet sniffing, DDOS, firewall.

I. INTRODUCTION

The exponential rise in the use of computer networks and digital technologies has resulted into difficulties with maintaining network availability, integrity, and confidentiality. The safeguarding of data and its associates in a network-based computer system possess a daunting challenge in the world today. With the ever-increasing digital traffic, hackers and malicious users' new ways of penetration and network intrusion take place. Network intrusion has increased following the never settling growth of internet technologies over different domains of social networking, online learning, digital business, e-education, and digitalization. This has put the security of various organizations be it a large-scale multinational firm or a local startup, all are vulnerable and at risk. An (IDS (Intrusion Detection System)) is created that will check for attacks within a computer base online network, as well as monitor the network and protect it from intruders.

- ^[8] To carry out this research, the NSL KDD Test datasets were analyzed and used for training our model and setting the standards and parameters. Machine-learning algorithms such as Naïve Bayes, Support Vector Machines (SVM), Random Forest, and Decision Tree were applied to determine the best accuracy for our IDS-System by classifying these attacks as anomaly or non-anomaly. Analyzing huge

data traffic is the summit task of a detection system. PyShark is a python library that captures the network traffic for us and presents us with packets with various attributes coupled along with it. We then pass it through our model and get a predicted score for the current packet. If its score is beyond the set threshold, we have an intrusion or else we continue to scan the network.

A. Introduction to Intrusion Detection System (IDS)

An IDS (Intrusion Detection System) is a networking module, or we can even consider them a software application capable of checking the network for anomalous packets, and policy violations. Firewalls, system protection software, and anti-malware programs by themselves are not sufficient to protect the complete network from an external attack. An in-depth security measure must pinpoint suspicious traffic once it passes the firewall and enters the network. IDS is known as "the second line of defense" for detecting anomalous packets at the first physical and then network layer after passing through the firewall and other security appliances on the edges of the network. We have two prominent detection methods of IDS.

- Signature based Method – Based on the previous malicious attacks, the IDS system will compare the patterns in the current packets and if a match is found, we declare it as malicious.
- Anomaly based Method – The downside of the signature based was if the attack was a new one, which hasn't been previously recorded it can go undetected, so to overcome this we have the anomaly-based IDS system. Advance ML/AI algorithms are used to see if the incoming traffic is suspicious or not suspicious.

B. Designing an Anomaly based IDS.

Owing to an endless array and new forms of penetration, a predefined set of patterns and signatures to represent such attacks becomes impossible. [12]Signature-based systems cannot distinguish between normal and malicious packets for which it does not have any signatures or patterns. So, to develop our system and to handle newer attacks, we have worked with anomaly-based detection techniques. So, we must train our anomaly-based detection system using some machine learning algorithm or deep learning technique, or AI-based algorithm. We have two stages in this process, the

training phase (we teach what regular-based behavior is) and the testing phase (current network traffic is compared with the regular behavior as per our data). The classifier will then use a mathematical model to check for any deviation from what is called a regular behavior and then accordingly will generate an alert.

II. FOR WHOM IS THIS SYSTEM

This System is designed to protect various organizations and institutes from various attacks.

- Big companies dealing with loads and loads of data and public information
- Schools, colleges, universities holding sensitive student information. Competitive exams papers all on their servers
- Healthcare systems
- Banking and cryptocurrency sector
- Private networks and crucial services

A. Objective

To create a software to sniff incoming packets

To collect the incoming packets and categorize them as safe or unsafe

To use advance machine learning algorithms to classify the traffic

Obtain a good accuracy in predicting the incoming packets

Create a user-friendly interface and design

Notify users in case of an attack

B. Scope

Network Intrusion Detection can be used to protect a given private or open network from attackers present on the worldwide network. This NIDS (Network Intrusion Detection System) can be employed in universities to secure papers and student confidential data, and Healthcare systems to add another level of security to prevent tampering or extraction of patient records.

C. Proposed System

The proposed model encompasses use of algorithms like Support Vector Machine (SVM), NaiveBayes, RandomForest, and DecisionTree.

Main aim behind the proposed model is to extract the common features between the NSL-KDD dataset and packet sniffer and train those features for real-time intrusion detection.

There will be transfer of packets at various network interfaces when using any kind of transfer (data, file etc.)

Using Packet Sniffer, real-time sniffing of packets takes place during these exchanges, whose features go through our intrusion detection system. We extract 7 out of the 42 features using the Pyshark python library by scanning all incoming traffic from the network interface. We can identify the IP address, the type of packet, and the current services being used. These features are passed over to our trained

model which classifies the packets and if there are any anomalous packets, it will notify the user as shown in Fig 1.

Using this we plan to identify if the packet is normal or abnormal based on the 7 features and then ascertain ourselves by seeing if its behavior is different from what is considered normal.

There will be 2 classes: normal and abnormal. Abnormal can further be classified as DoS, U2R, Probe, R2L attacks with increased accuracy.

III. SYSTEM DESIGN AND ARCHITECTURE

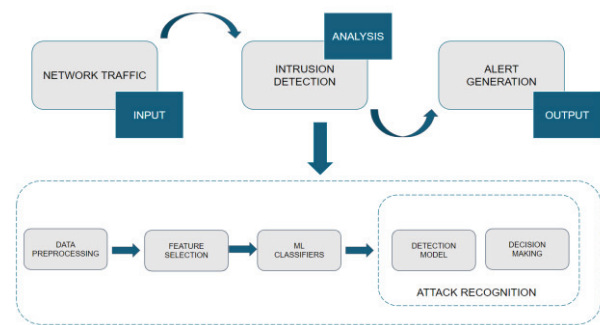


Fig. 1. System Architecture of an Intrusion Detection System

System design of this Intrusion System demonstrates a relation with the deployed model and incoming packets and the alert generation system. The basic interaction between the user and the product, and how the packets are classified based on real world analysis is shown.

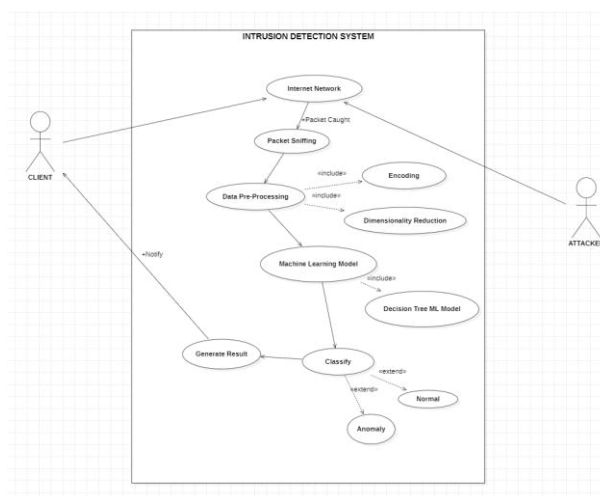


Fig. 2. IDS Use Case Diagram

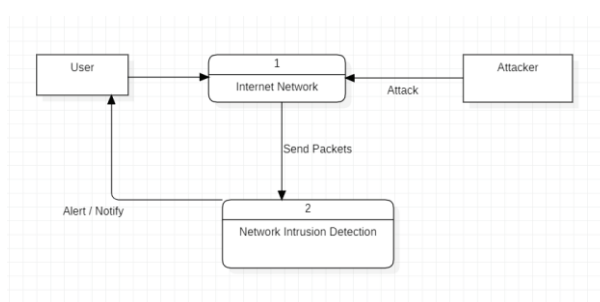


Fig. 3. IDS Level 0 DFD Diagram

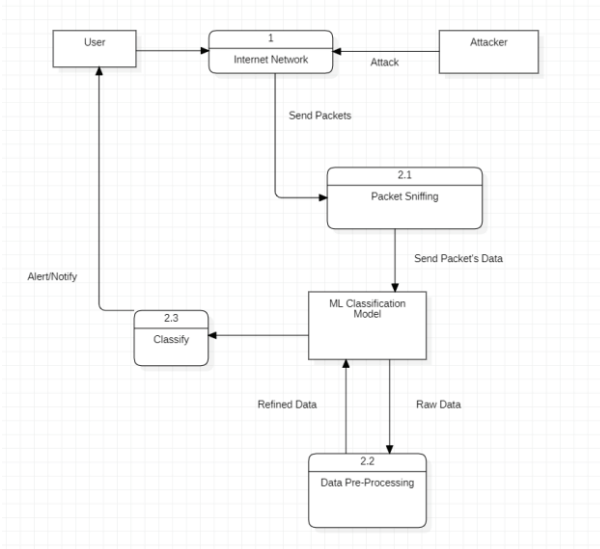


Fig. 4. IDS Level 1 DFD Diagram

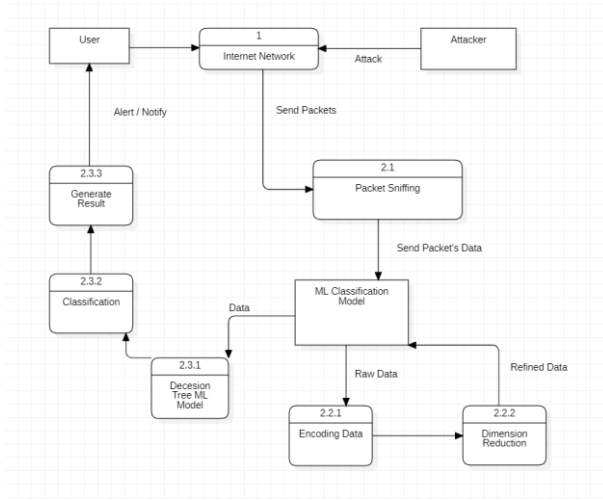


Fig. 5. IDS Level 2 DFD Diagram

IV. USING MACHINE LEARNING ALGORITHMS

We need to successfully classify the [6] packets based on the deviation the current data has with respect to the train data that is the NSL – KDD dataset.

Proposed model consists of using algorithms like SVM, naïve-bayes, random-forest, logistic regression and decision tree.

A. RandomForest Regressor

Random Forest (RFR) is a supervised algorithm and makes use of ensemble methods for classification along with regression. (RFR) builds multiple decision trees and once done combines or merges them and acquires a more precise and stable prediction. Each Tree in RFR is trained individually and consists of random data. This degree of randomness enables us to make the model more robust compared to a standalone decision tree, and unlikely to overfit on the training data. Such a model can recursively build upon the various nodes by taking into consideration the features of the incoming packets and based on the constraints

over the NSL KDD dataset branch out and obtain the corresponding score.

TABLE I. RANDOM FOREST ACCURACY MODEL TABLE

MOEDLS	ACCURACY	RMS	F1 SCORE	RECALL	PRECISION
RANDOM FOREST	0.9911	0.094	0.9917	0.9944	0.9889

B. Support Vector Machine

An SVM depicts supervised ML algorithm. SVMs (Support Vector Machine) draw from the notion of narrowing down to a hyperplane which perfectly separates a dataset into two classes. [8]S.V are data points in proximity with hyperplane, the coordinates of a dataset, once excluded, would amend a considered position upon dividing hyperplane. Due to this, they are rightly determined as the critical elements within data set. Using this division of the hyperplane we can separate the anomalous packets from the non-anomalous and try for a better precision score.

TABLE II. SUPPORT VECTOR MACHINE ACCURACY MODEL TABLE

MOEDLS	ACCURACY	RMS	F1 SCORE	RECALL	PRECISION
SVM	0.6159	0.6196	0.4602	0.3067	0.9217

C. Naïve Bayes

A frequently used, ML classifier also depicted as a probabilistic classifier. We can define it with a quite simple Bayesian network. Naïve-Bayes classifiers are characterized for classification and have been a prominent solution to problems like spam detection and malicious activity identification. [4]It performs well also with high-dimensional data points alongside a dynamic range of data points, making it a strong prospect for the NSL KDD dataset.

$$P(A/B) = (P(B/A)*P(A)) / P(B)$$

TABLE III. NAÏVE BAYES ACCURACY MODEL TABLE

MOEDLS	ACCURACY	RMS	F1 SCORE	RECALL	PRECISION
NAIVE BAYES	0.874	0.3548	0.8897	0.9515	0.8354

D. Decision Tree

A Decision tree is regarded by the conventional notion as a flowchart-based tree structure, in which the nodes present within the end constraints depicts a test performed on certain attribute under consideration, branch depicts a result of the test, which was performed. Using this we can recursively build our model with the NSL KDD regular patterns and branch from the initial start point towards the terminals.

TABLE IV. DECISION TREE ACCURACY MODEL TABLE

MOEDLS	ACCURACY	RMS	F1 SCORE	RECALL	PRECISION
DECISION TREE	0.9911	0.094	0.9917	0.9924	0.9909

E. Logistic Regression

[12]Used to calculate the probabilistic outcome of a said class or event taking place. This can be used to calculate the individual probabilities of every feature in the dataset and based on certain established threshold the characteristics will

be determined and prediction if the network has an intrusion or no will be made.

TABLE V. LOGISTIC REGRESSION ACCURACY MODEL TABLE

MOEDLS	ACCURAC Y	RMS	<i>FI SCOR E</i>	<i>RECAL L</i>	<i>PRECISIO N</i>
LOGISTIC REGRESSIO N	0.9085	0.302 5	0.9156	0.9293	0.9023

V. USING PYSHARK FOR PACKET SNIFFING

Pyshark is a python library that uses Wireshark's dissectors to parse packets received on the network interface. There are other various parsers, but this has the best efficiency because Pyshark does not capture packets on its own but uses Wireshark's ability to export the files (XMLs) and uses those files to parse. Depending upon the situation, Pyshark provides "Capture" objects such as Live, Remote, File, and InMenu. Pyshark creates an iterator after reading the files and is used to pick out packets. We can also combine Pyshark objects with various filters to obtain desired data packets.

VI. NSL KDD DATASET

^{[13][11]}NSL-KDD is an optimized dataset created from the KDD'99 dataset after removing some known problems that create difficulties in applying the dataset. The advantage being it is more effective in comparing the different intrusion detection methods for the researchers. This new dataset makes it possible to train the model on the entire dataset rather than just a portion. Overall, this will help to create a uniform benchmark for any further research.

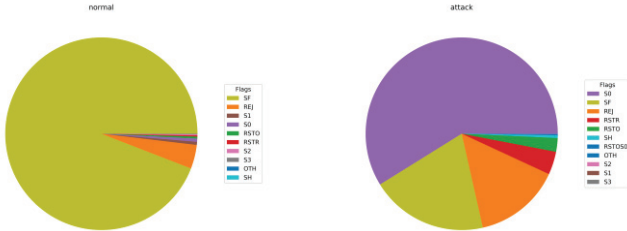


Fig. 6. NSL KDD dataset analysis

VII. FEATURE SELECTION AND ANALYSIS

From the NSL KDD Dataset and the packet's data sniffed by the sniffer. We have found 7 common features between the dataset and the packet's data, which would be considered to be worked upon our training along with our testing of ML model. On a whole 41 features are present and we are using only 7 features. These are:-

1. Protocol
2. Land
3. Urgent
4. Count.
5. Srv count.
6. Dst_host_count
7. Dst_host_srv_count.
- 8.

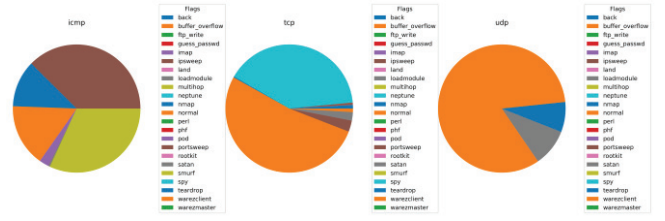


Fig. 7. Protocols and Feature selection.

VII. INTRUSION DETECTION AND PRESENTATION

^[7]The user needs to first authorize this software then we will start analyzing the packets based on the model we have trained. So, we begin by sniffing packets, then we normalize the data and perform dimensionality reduction, and once all this is done and the data is clean and pure, we check for certain features of the threshold limit and then classify them into various categories such as normal, DOS, R2L, U2L and more. Once we do all this, we generate the result and then display it to the user.

- A. System functioning and result analyses
 - a. Providing the user 3 options
 - i. Check for anomaly
 - ii. Check Captured Packets
 - iii. Exit
- B. Database connectivity.
- C. Graphical User Interface for result display.

VIII. LATEST DEVELOPMENT

The, latest research and development make use of "Deep Neural Network (DNN) this model was trained from the NSL-KDD using 28 features". ML pipeline gives categorical encoding of data in a sequential format along with feature scaling, after which it transmits the real time data to trained DNN model and ends up making predictions. Such complex algorithms require lots of training and well-defined parameters while designing the hidden layers and biasing. The amount of research in this field based on hidden networks is quite limited and further research is being carried out. The artificial neural network is the architecture design and inspired from the Human brain, which leads towards the development of RT-IDS systems.

IX. CONCLUSION

Thus, we have successfully sniffed the network for packets and carefully selected the protocols for establishing whether the packets are malicious or not using anomaly-based network intrusion with the help of the decision tree algorithm against the NSL-KDD dataset. This system can be easily used on individual systems or within the intranet of a given organization and sniff the network for attacks.

Future studies on this topic can be to develop a new database that would eliminate the deficiency of lacking features in the current framework.

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